Protocolos de comunicaciones de alta velocidad

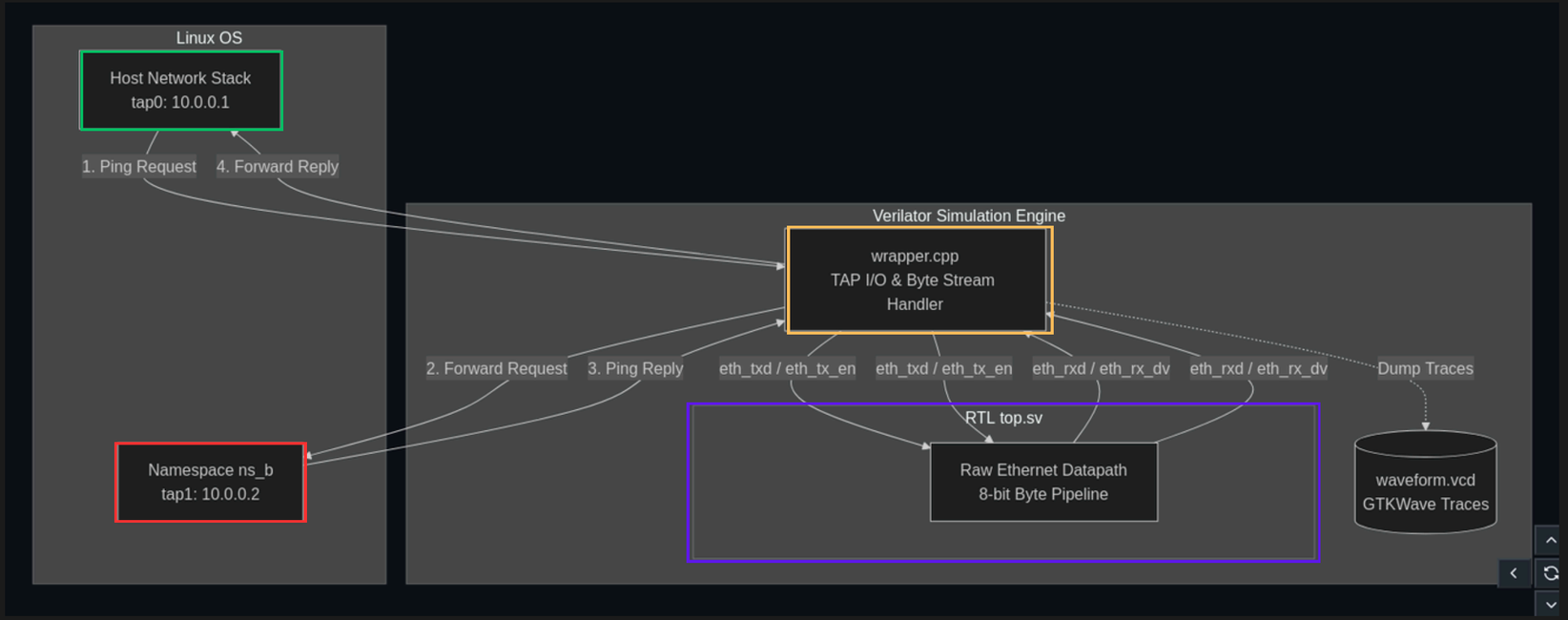
Ethernet-phy

Lab1 Raw Ethernet

Fundación Fulgor

Facundo Guerrero

Architecture Setup



- Consists of two virtual interfaces, Tap0 and Tap1, which can communicate with each other (for example, using the *ping* command).
- In between them, there is a *wrapper* that takes the data and processes it according to the specific case being studied on each laboratory. Then injects the data into the input pins of the *top.sv* circuit. At the same time, it records a *.vcd* file for waveform visualization.
- The digital circuit *top.sv* is a module used to simulate the hardware. It assigns the signals and, when necessary, creates debug signals.
- tcpdump is used in parallel to generate a *.pcap* file for subsequent analysis.
- In this case, Lab 1, the data goes directly (raw) from one TAP interface to the other.
- Finally, the packets are analyzed using Wireshark and GTKWave.

Architecture Setup

```
~/Documentos/fulgor/labs/ethernet-phy-labs/scripts  main
❯ sudo ./setup_netns.sh
[sudo] contraseña para facusu:
═══ 1. Limpiando configuración previa ═══
═══ 2. Creando interfaces TAP y Namespace de red ═══
═══ 3. Configurando Direcciones IP y Estado ═══
═══ 4. Deshabilitando Offloading para Simulación RTL ═══
═══ 5. Verificando estado del entorno ═══
→ Interfaz tap0 (Host):
5: tap0: <NO-CARRIER,BROADCAST,MULTICAST,PROMISC,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether 5a:f4:04:87:86:e7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.1/24 scope global tap0
       valid_lft forever preferred_lft forever
→ Interfaz tap1 (Namespace ns_b):
6: tap1: <NO-CARRIER,BROADCAST,MULTICAST,PROMISC,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether ce:e0:bc:43:fd:85 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.2/24 scope global tap1
       valid_lft forever preferred_lft forever
================================================
Entorno preparado con éxito. Listo para la simulación.
================================================

~/Documentos/fulgor/labs/ethernet-phy-labs/scripts  main
❯
```

1. We run *setup_netns.sh* to create the TAPS

```
~/Documentos/fulgor/labs/ethernet-phy-labs/lab1-raw-ethernet  main
❯ verilator --cc --trace --exe --build -j 0 -Wall -Wno-DECLFILENAME wrapper.cpp top.sv -o emulator

make: se entra en el directorio '/home/facusu/Documentos/fulgor/labs/ethernet-phy-labs/lab1-raw-ethernet/obj_dir'
g++  -I.  -MMD -I/usr/share/verilator/include -I/usr/share/verilator/include/vltstd -DVERILATOR=1 -DVM_COVERAGE=0 -DVM_SC=0 -DVM_TIMING=0 -DVM_TRACE=1 -DVM_TRA
DVM_TRACE_SAIF=0 -DVM_VPI=0 -faligned-new -fcf-protection=none -Wno-bool-operation -Wno-int-in-bool-context -Wno-shadow -Wno-sign-compare -Wno-subobject-linka
 -Wno-uninitialized -Wno-unused-but-set-parameter -Wno-unused-but-set-variable -Wno-unused-parameter -Wno-unused-variable      -Os  -c -o wrapper.o ../wrapper.
g++ -Os  -I.  -MMD -I/usr/share/verilator/include -I/usr/share/verilator/include/vltstd -DVERILATOR=1 -DVM_COVERAGE=0 -DVM_SC=0 -DVM_TIMING=0 -DVM_TRACE=1 -DV
=1 -DVM_TRACE_SAIF=0 -DVM_VPI=0 -faligned-new -fcf-protection=none -Wno-bool-operation -Wno-int-in-bool-context -Wno-shadow -Wno-sign-compare -Wno-subobject-l
pare -Wno-uninitialized -Wno-unused-but-set-parameter -Wno-unused-but-set-variable -Wno-unused-parameter -Wno-unused-variable      -c -o verilated.o /usr/shar
d.cpp
g++ -Os  -I.  -MMD -I/usr/share/verilator/include -I/usr/share/verilator/include/vltstd -DVERILATOR=1 -DVM_COVERAGE=0 -DVM_SC=0 -DVM_TIMING=0 -DVM_TRACE=1 -DV
=1 -DVM_TRACE_SAIF=0 -DVM_VPI=0 -faligned-new -fcf-protection=none -Wno-bool-operation -Wno-int-in-bool-context -Wno-shadow -Wno-sign-compare -Wno-subobject-l
pare -Wno-uninitialized -Wno-unused-but-set-parameter -Wno-unused-but-set-variable -Wno-unused-parameter -Wno-unused-variable      -c -o verilated_vcd_c.o /us
rilated_vcd_c.cpp
g++ -Os  -I.  -MMD -I/usr/share/verilator/include -I/usr/share/verilator/include/vltstd -DVERILATOR=1 -DVM_COVERAGE=0 -DVM_SC=0 -DVM_TIMING=0 -DVM_TRACE=1 -DV
=1 -DVM_TRACE_SAIF=0 -DVM_VPI=0 -faligned-new -fcf-protection=none -Wno-bool-operation -Wno-int-in-bool-context -Wno-shadow -Wno-sign-compare -Wno-subobject-l
pare -Wno-uninitialized -Wno-unused-but-set-parameter -Wno-unused-but-set-variable -Wno-unused-parameter -Wno-unused-variable      -c -o verilated_threads.o /
verilated_threads.cpp
python3 /usr/share/verilator/bin/verilator_includer -DVL_INCLUDE_OPT=include Vtop.cpp Vtop___024root__0.cpp Vtop__Trace__0.cpp Vtop___024root__Slow.cpp Vtop__
Syms__Slow.cpp Vtop__Trace__0__Slow.cpp Vtop__TraceDecls__0__Slow.cpp > Vtop__ALL.cpp
g++ -Os  -I.  -MMD -I/usr/share/verilator/include -I/usr/share/verilator/include/vltstd -DVERILATOR=1 -DVM_COVERAGE=0 -DVM_SC=0 -DVM_TIMING=0 -DVM_TRACE=1 -DV
=1 -DVM_TRACE_SAIF=0 -DVM_VPI=0 -faligned-new -fcf-protection=none -Wno-bool-operation -Wno-int-in-bool-context -Wno-shadow -Wno-sign-compare -Wno-subobject-l
pare -Wno-uninitialized -Wno-unused-but-set-parameter -Wno-unused-but-set-variable -Wno-unused-parameter -Wno-unused-variable      -c -o Vtop__ALL.o Vtop__ALL
g++    wrapper.o verilated.o verilated_vcd_c.o verilated_threads.o Vtop__ALL.a    -pthread -lpthread -latomic   -o emulator
rm Vtop__ALL.verilator_deplist.tmp
make: se sale del directorio '/home/facusu/Documentos/fulgor/labs/ethernet-phy-labs/lab1-raw-ethernet/obj_dir'
- V e r i l a t i o n   R e p o r t: Verilator 5.050 2026-07-01 rev v5.050
- Verilator: Built from 0.027 MB sources in 2 modules, into 0.031 MB in 8 C++ files needing 0.000 MB
- Verilator: Walltime 6.998 s (elab=0.014, cvt=0.027, bld=6.950); cpu 0.018 s on 4 threads; allocated 346.449 MB

~/Documentos/fulgor/labs/ethernet-phy-labs/lab1-raw-ethernet  main
❯ sudo ./obj_dir/emulator
[TAP] Interfaz tap0 vinculada correctamente (host).
[TAP] Interfaz tap1 vinculada correctamente (ns_b).
[EMULADOR LAB 1] Ejecutando simulación...
```

2. We use verilator to compile and generate an executable taking wrapper and top.sv files. This create an emulation executable

```
~
❯ ip addr show dev tap0
5: tap0: <BROADCAST,MULTICAST,PROMISC,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 5a:f4:04:87:86:e7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.1/24 scope global tap0
       valid_lft forever preferred_lft forever
    inet6 fe80::58f4:4ff:fe87:86e7/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever

~
❯ sudo ip netns exec ns_b ip addr show dev tap1
[sudo] contraseña para facusu:
6: tap1: <BROADCAST,MULTICAST,PROMISC,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether ce:e0:bc:43:fd:85 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.2/24 scope global tap1
       valid_lft forever preferred_lft forever
    inet6 fe80::cce0:bcff:fe43:fd85/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever

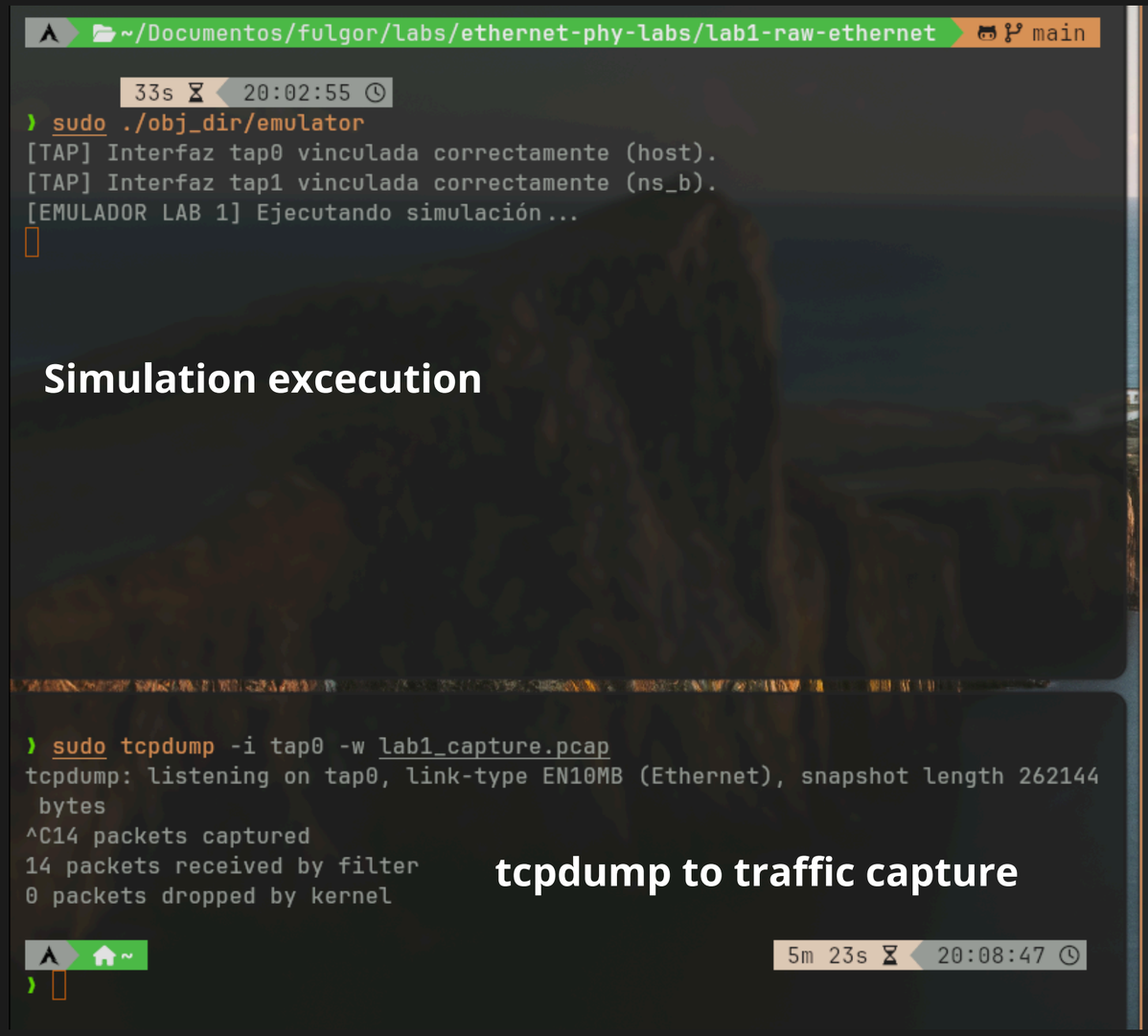
~
❯
```

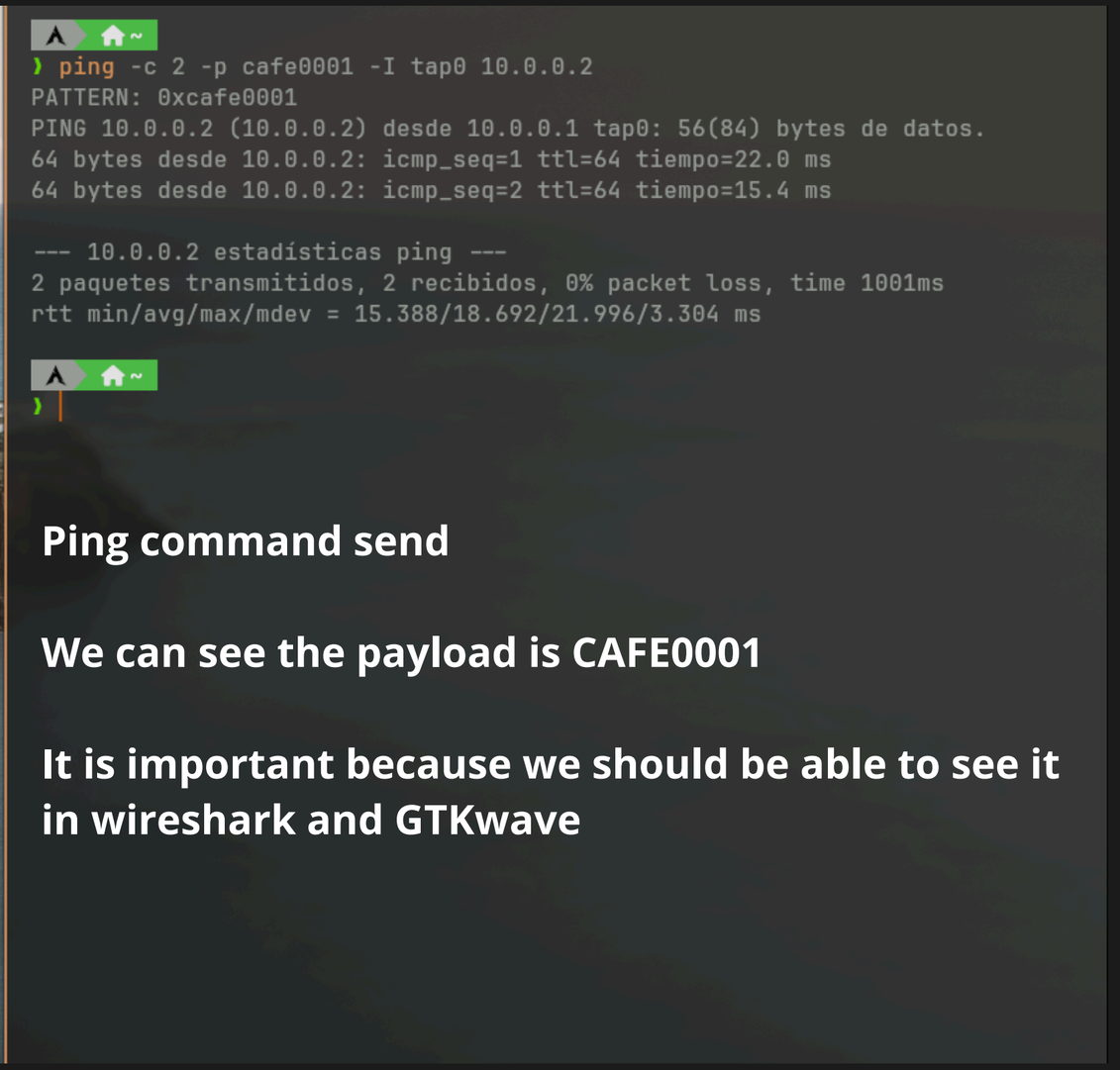
3. Finally there are the two taps and its two MACs addresses and ips

TAP	IP	MAC
Tap0	**10.0.0.1**	**5a:f4:04:87:86:e7**
Tap1	**10.0.0.2**	**ce:e0:bc:43:fd:85**

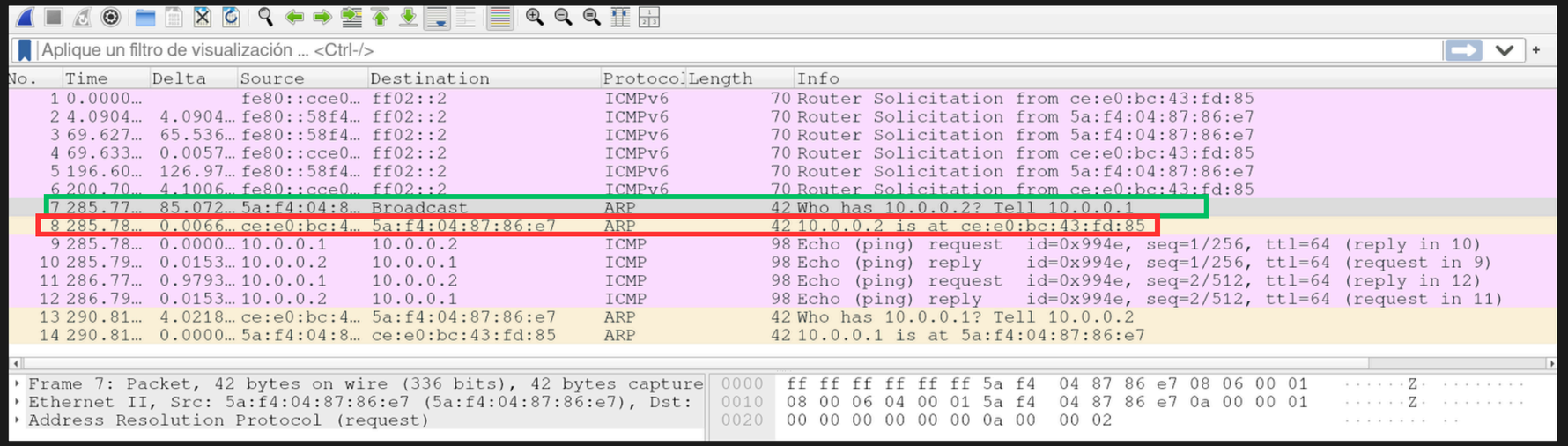
Execution

It is executed simultaneously in three terminals.

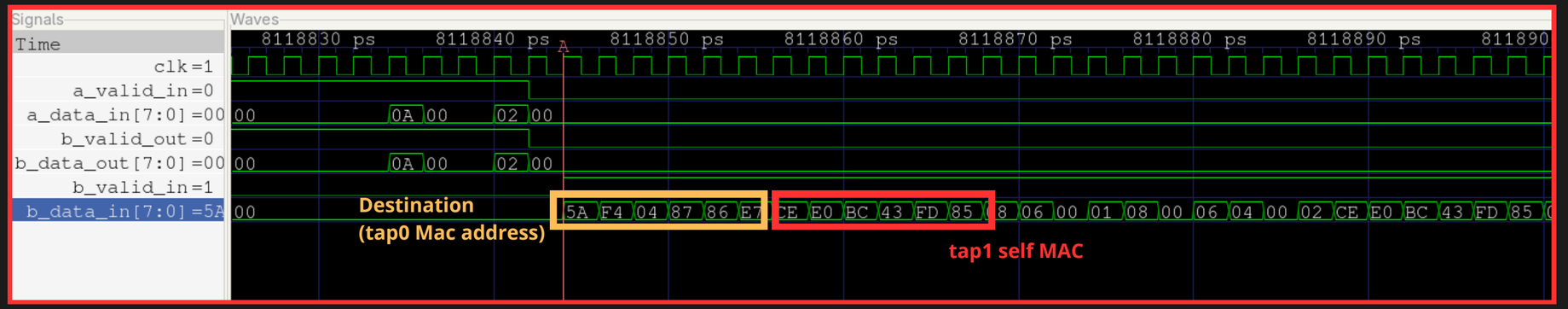




Analysis







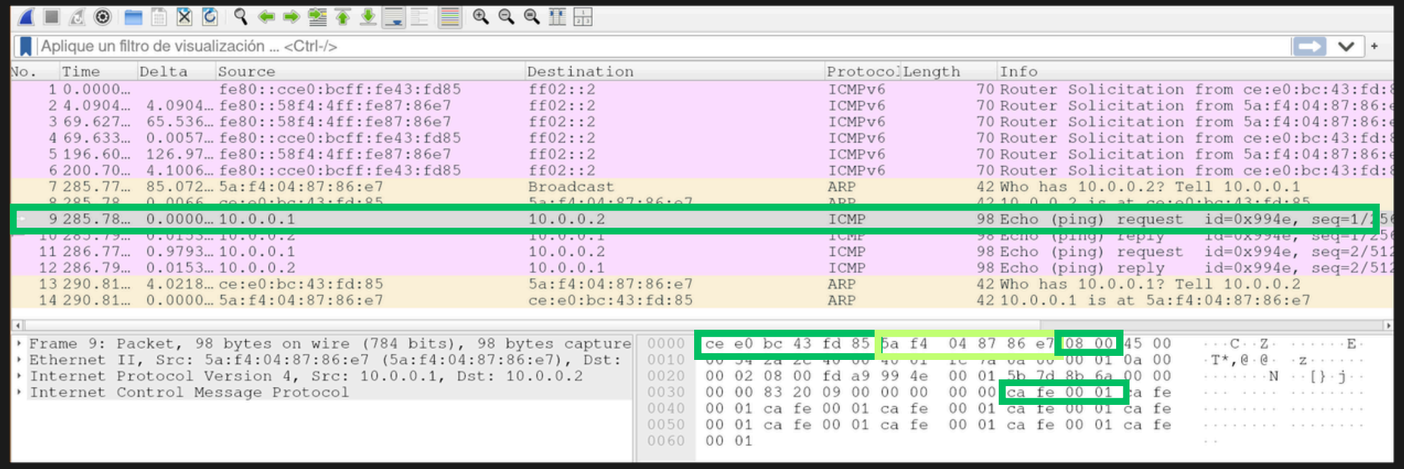
The first packet we analize is the Broadcast.

Here Tap0 Start with an ARP packet asking for the IP.

Then Tap1 responds with its own MAC direction

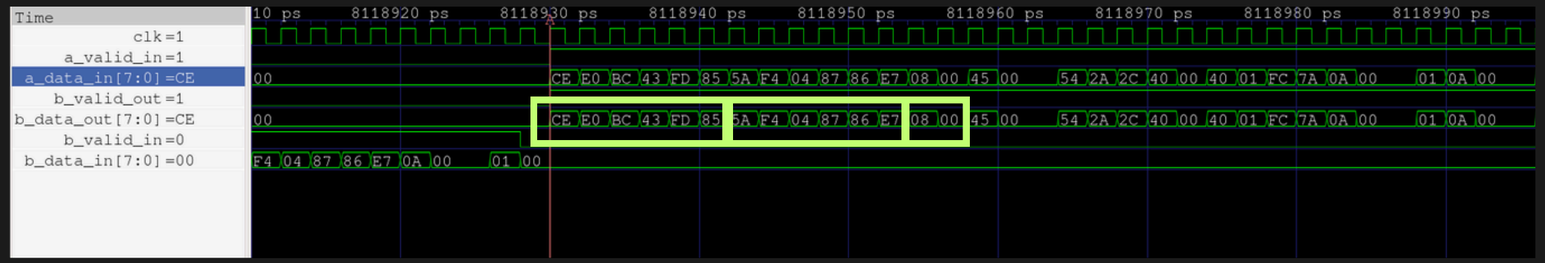
Analysis

Now we analyze the packet of interest, the Ping Request from Tap0 to Tap1.

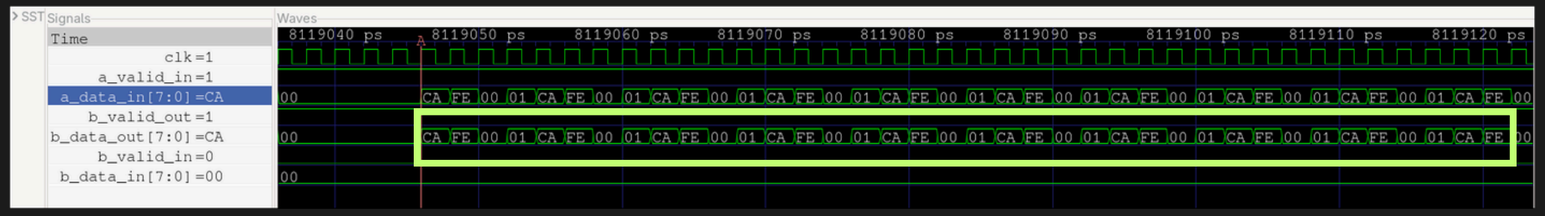


The packet starts with tap1 MAC address because is the destination. Then MAC source (tap0 MAC) and ethertype

And we can see the payload CAFE0001 that we defined previously



Starts with tap1 MAC
After Mac Destination
And EtherType 0800 = ipv4

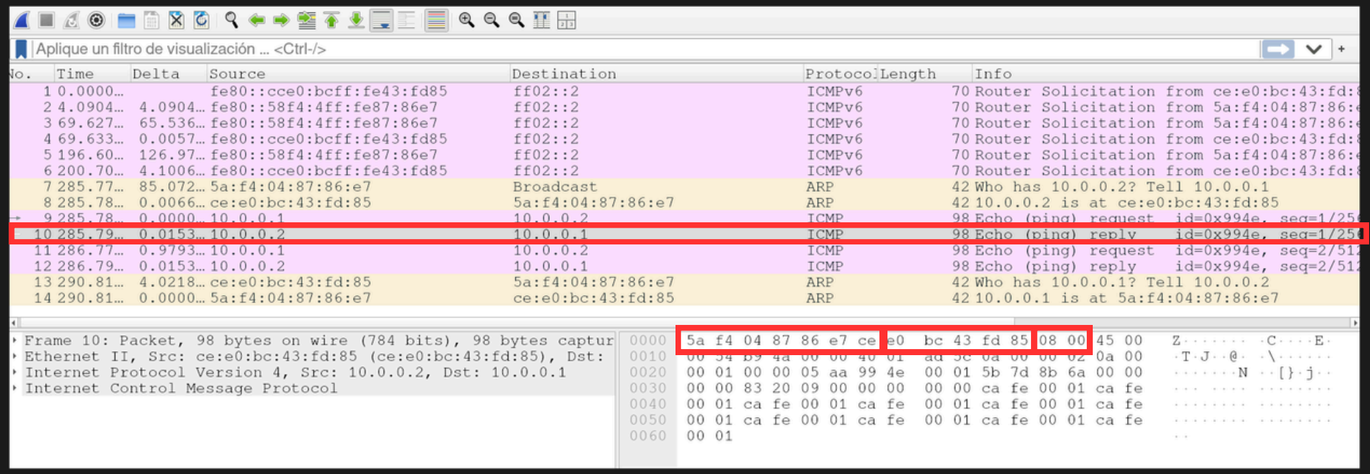


CAFE0001 Payload

That means that the packets are being sent correctly.

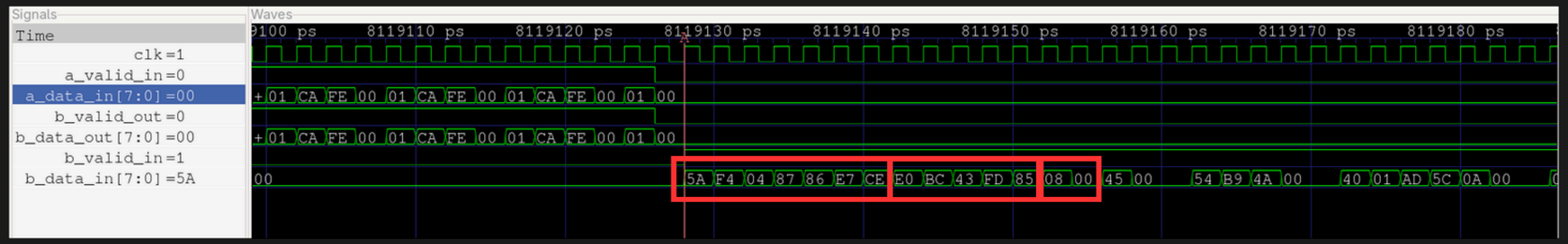
Analysis

Finally we analyze the Ping reply from tap1 to tap0

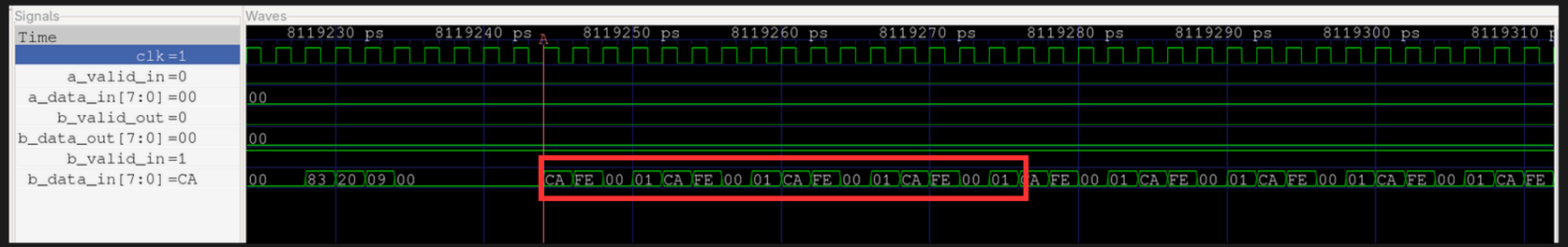


The packet starts with tap0 MAC address because is the destination. Then tap1 source MAC and Ethertype

And we can see the payload CAFE0001 that we defined previously



Starts with tap0 MAC After Mac Destination And EtherType 0800 = ipv4



CAFE0001 Payload

The packets are sending from tap0 to tap1 and vice versa correctly